

WHEN IMMEDIATE AGREEMENT CANNOT IDENTIFY PERSISTENT PREFERENCE

Anonymous research draft
Empirical completion pending

ABSTRACT

Learning from a user’s next message requires specifying what that message measures. Immediate agreement may reflect a transient change in expression or a persistent change in preference. We give an exact finite-state construction in which these mechanisms induce the same historical observations, including observed baseline states, but rank population actions differently under a declared post-action matching objective. For a new user whose baseline state is unavailable, the two-world expected-risk minimax regret is $2/15$. The result does not apply unchanged when the deployment user’s baseline state is observed, and post-action matching is not itself a welfare criterion. Known, informative delayed measurements identify the finite transition law under explicit measurement assumptions. Existing constructed simulations show both benefits and limitations of standard residual correction: a simpler anchor estimator can outperform its propensity-weighted version in policy regret. We specify a prospective neural comparison with equal delayed-label access, direct predictive endpoints, and semantic and acquisition controls. **The neural study and external persistence validation have not been run; this draft establishes neither neural method superiority nor real assistant-induced preference change.**

1 INTRODUCTION

The next user message provides inexpensive supervision for an assistant, but it is also a response to the assistant’s own action. Hindsight self-distillation conditions a teacher on that subsequent message and trains a student that did not observe it (Kleine Buening et al., 2026). Such training can be useful without making any causal claim. A stronger claim—that the update improves a persistent user preference—requires a connection between the observed utterance and a specified persistent target. That connection cannot be inferred merely because the user agrees with the preceding response.

Consider a recommendation followed by assent. The user may temporarily echo the recommendation while retaining an earlier preference, or the recommendation may change a preference that remains after the immediate conversational context is removed. If both mechanisms generate the same observable logs, collecting more of those logs cannot distinguish them. This ambiguity concerns the measurement of latent state, in addition to familiar uncertainty about the function used to predict an observed label. A delayed probe helps only when its interpretation is justified independently: delay alone does not make a report neutral, stable, or nonreactive.

This problem sits close to several established lines of work. SLIFT decomposes feedback for selective learning (Li et al., 2026); PUMA models action-conditioned user dynamics and inferred state (Luo et al., 2026). Work on privileged likelihood already separates loss or gradient fidelity from utility (Nguyen et al., 2026). Human preference elicitation can change reported preferences under a fixed underlying reward (Hatgis-Kessell et al., 2026), and influenceable reward functions raise competing normative objectives (Carroll et al., 2024). Our remaining question is specifically whether independent persistence measurements resolve a consequential ambiguity for interaction learning. A generic causal-personalization framing or a renamed residual estimator would not establish a new contribution.

The present draft separates three objects. First, an exact witness establishes observational nonidentification under a restricted decision setting. Second, constructed finite-state simulations evaluate

ordinary difference estimation; they do not measure how frequently the ambiguity occurs among people. Third, a prospective neural screen asks whether correction improves prediction of stipulated delayed labels beyond simple learners receiving the same labels. That predictive experiment is not an empirical test of the theorem’s action-induced state objective. Its completion and an externally defensible measurement study remain necessary for the intended paper.

2 OBSERVED FEEDBACK AND DECLARED TARGETS

Let Z_0 be a baseline persistent state, A an assistant action, $Z_1(A)$ the persistent state after that action, and S_1 a transient expressed stance. The observed next message O is emitted from the latent state. In the exact witness, the emission is the same in both mechanisms and simply reports S_1 . Historical data contain (Z_0, A, O) ; thus the witness does not rely on hiding the historical baseline state. The logging action has positive probability under each baseline state.

We distinguish the value of an action that changes the target state,

$$V_m(a) = \mathbb{P}_m\{a = Z_1(a)\}, \quad m \in \{E, T\}, \quad (1)$$

from the value of a predictive distribution for a fixed delayed label B ,

$$U(\pi) = \mathbb{E}[\pi(B \mid X)]. \quad (2)$$

In Equation 1, the evaluated action can change the state that supplies its own reward. In Equation 2, the evaluation label is fixed when the prediction is scored. Optimizing either quantity is not automatically equivalent to protecting autonomy, consent, pre-treatment preferences, or long-term welfare. Those require separately defended objectives.

In the neural screen, B is a constructed delayed-expression label on a held-out task from an observed user cohort. The prediction is the full-vocabulary probability of the native token for the correct option; other vocabulary tokens receive zero credit. If $M(X)$ is the total probability of the four answer tokens and $q(B \mid X)$ their normalized conditional probability, then $\pi(B \mid X) = M(X)q(B \mid X)$. Consequently, both full-vocabulary NLL and correct-token probability can improve solely by placing more mass on the answer format. We report and gate conditional semantic correctness separately.

3 A RESTRICTED IDENTIFICATION WITNESS

Set $\mathbb{P}(Z_0 = 1) = 3/5$ and log actions independently with probability $1/2$. When an action conflicts with Z_0 , let the user copy action zero with probability one and action one with probability zero. In mechanism E , copying changes expression only; in mechanism T , it changes persistent state as well. The resulting immediate observation is $O = 0$ when $A = 0$ and $O = Z_0$ when $A = 1$ in both mechanisms. Thus their complete historical laws are identical.

The action values under Equation 1 are

Mechanism	$V(0)$	$V(1)$
Expression only E	$2/5$	$3/5$
Persistent transition T	1	$3/5$

For a new anonymous user whose Z_0 is unavailable to the policy, let q be the learner’s overall probability of action one, integrating over historical data and any randomization. Identical historical laws imply the same q in both worlds. The expected regrets are $(1 - q)/5$ and $2q/5$, so their maximum is minimized at $q = 1/3$, with regret $2/15$. This statement covers data-dependent deterministic rules as well as randomized rules. A bound of .2 for every deterministic learner would be false because the random historical sample itself supplies randomization; Appendix A gives a counterexample.

The deployment restriction matters. If the current user’s Z_0 is observed, the personalized rule $A = Z_0$ achieves value one in both witness worlds. The historical mechanisms remain nonidentified, but this particular decision problem no longer incurs regret. The construction therefore establishes neither a universal personalization lower bound nor inevitable harm from adaptation.

A stipulated neutral probe maps (Z_1, S_1) to (Z_1, Z_1) without changing Z_1 . Adding its observation gives total variation distance $3/10$ between the witness laws. More generally, with a known state-emission matrix M of full row rank, conditional probe distributions satisfy $Q_a = T_a M$ and identify $T_a = Q_a M^\dagger$. This is standard linear identification, not a new general matrix theorem. Unknown emission, missing action support, ill conditioning, state-label ambiguity or probe-induced transitions change the conclusion. Complete assumptions and proofs are in Appendix A.

4 STANDARD RESIDUAL CORRECTION AND FAIR INFORMATION

For a fixed model parameter θ , let $g_i^I(\theta)$ be a proxy loss gradient based on immediate feedback and $g_i^D(\theta)$ the corresponding gradient under a valid delayed label. With N immediate examples and a uniform sample of n paired delayed anchors, ordinary difference estimation uses

$$\hat{g}(\theta) = \frac{1}{N} \sum_{i=1}^N g_i^I(\theta) + \frac{1}{n} \sum_{i \in \mathcal{A}} \{g_i^D(\theta) - g_i^I(\theta)\}. \quad (3)$$

For fixed θ and the stated sampling design, the correction targets the full delayed-gradient mean. Nonuniform selection requires the appropriate known inclusion probabilities and support. This estimator belongs to classical difference estimation and prediction-powered inference (Angelopoulos et al., 2023); the arithmetic is not the proposed novelty.

The fixed-parameter identity does not establish a neural training guarantee. Our SDPO-style teacher depends on the current adapter, is detached within an update, and changes between updates. Reusing a fixed anchor set while adapting parameters does not preserve an unconditional fixed-parameter unbiasedness claim along the entire trajectory. Clipping a correction, clipping gradients, finite optimization and selecting an action from an estimated value further separate estimation bias from decision regret. Even an unbiased value estimate need not induce a better decision than a biased estimate with favorable ranking.

Every sparse comparator must receive the same available immediate logs, delayed labels and meta-data. A full-delayed oracle is an extra-information acquisition control, not an equal-budget baseline. Equal optimizer updates also do not imply equal computation: the residual arm needs additional teacher and student forwards. We report label access and computation separately and require matched-compute comparisons before claiming method superiority.

5 EXISTING CONSTRUCTED EVIDENCE

The retained finite-state studies are developmental experiments on designed cells. We recomputed their displayed aggregates directly from saved cell rows and retained every relevant comparator. The first study uses 150 qualifying design cells at each anchor budget, with 2,000 coupled synthetic draws per cell. At 16 anchors per action, mean expression regret is .08628067 for immediate feedback, .01377017 for ordinary anchors and .00934317 for clipped augmentation. At 64 anchors the latter two values are .00431483 and .00268983. These improvements are properties of this constructed screen, not neural or human effect estimates.

The robustness screen is more cautionary. Its clean ranking-reversal subset contains 13 designed configurations per selection level. At random selection, ordinary anchors have regret .00380385, better than augmented IPW at .00417692, while unweighted augmentation is better than both at .00222308. At the strongest included selection bias, ordinary anchors attain .00000769 versus .00218077 for augmented IPW. The original bias-correction checks still pass. The conclusion is that bias control and policy superiority are distinct, and that omitting a simple comparator changes the interpretation. Figure 1 shows all comparators rather than only the favorable pair.

A separate finite longitudinal calculation also rejects an automatic-harm interpretation: symmetric adaptation need not increase mean baseline drift. These retained results motivate sharper measurement and decision questions; they do not jointly establish a successful learning method. Original protocols, raw rows, replay receipts and corrected claim classifications remain preserved.

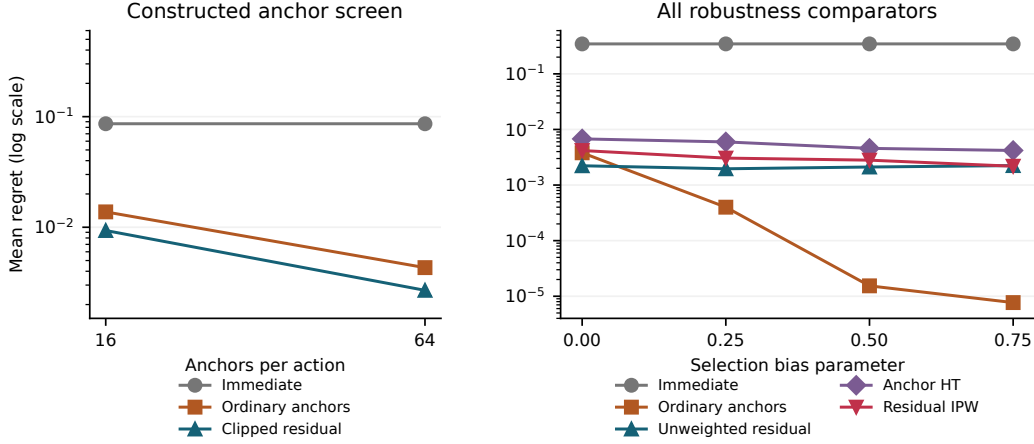


Figure 1: Existing constructed simulations, recomputed from saved design-cell rows. Left: 150 qualifying cells per budget. Right: all five comparators in 13 clean ranking-reversal cells per selection level. The logarithmic regret axis preserves the raw comparator. Cells and coupled draws are not independent human samples. Ordinary anchors can beat augmented IPW; unweighted augmentation is best at random selection in this grid.

Arm	Signal	Delayed bases	DEV result
Raw	Immediate SDPO-style loss	0	<i>Not run</i>
Full oracle	Delayed SDPO-style loss	630	<i>Not run</i>
Anchor SFT	Pooled native-label NLL	64	<i>Not run</i>
Anchor SDPO	Pooled delayed teacher	64	<i>Not run</i>
Residual	Immediate plus paired correction	64	<i>Not run</i>
Mixture	0.5 immediate +0.5 anchor SDPO	64	<i>Not run</i>

Table 1: Prospective comparison, not experimental results. All sparse arms may use the same 630 immediate logs and the same 64 delayed bases. A no-update baseline and fixed profile/memory readouts are additional controls.

6 PROSPECTIVE NEURAL SCREEN

Status and population. No result from the reduced screen exists. The prospective task uses 630 learning bases derived from PAHF and 96 development bases, with four option rotations per base. The learning cohort has 20 source-user display-name groups; all 19 development groups occur in learning. The prepared data do not provide a separate unique person identifier. We therefore target held-out tasks within this cohort, not unseen-user generalization. The 256 reserved confirmation bases remain unopened and are not consumed by preparation or development verification. The stipulated old/new labels and transition narratives are constructed; they do not measure assistant-caused human persistence.

Arms and budgets. Pool the existing four disjoint 16-base panels into one 64-base delayed-label budget shared by all sparse learners. Each arm is one Qwen3.5-9B model with the same immutable base revision, identical initial rank-eight adapter and a fresh AdamW optimizer. Use 35 fixed updates at learning rate 10^{-4} , no weight decay, and the fixed final checkpoint. The population schedule visits each learning base once, 18 rows per update. The anchor schedule presents two bases in all four rotations per update: all 64 bases in the first 32 updates, then six hash-selected revisits, totaling 280 anchor-row presentations.

The teacher loss is first-answer-token full-vocabulary $\text{KL}(p_\theta \| \text{stopgrad } p_\theta^{\text{hindsight}})$. This is an SDPO-style adaptation, not a reproduction of the released multi-token algorithm. Full-sequence SDPO and SLIFT given the same delayed information belong to the conditional follow-up, not to an unearned baseline claim in Table 1.

Qualification and outcomes. The exact downstream serialization first undergoes a learning-only 16-base, four-rotation, two-context interface check. Every context/label cell must have at least 14 of 16 correct conditional choices, mean target probability at least .70 and mean answer mass at least .10. The raw and full-oracle arms then run before any sparse arm. The oracle must improve old-target NLL by at least .10 over both no-update and raw; raw must improve new-target NLL by at least .10 over no-update. Every corresponding full-vocabulary and conditional semantic probability change must be positive. Failure is an invalid acquisition assay, not a causal negative.

If acquisition qualifies, the residual must improve old-target NLL by at least .03, full-vocabulary correct-choice probability by .02, and conditional semantic correctness by .01 against each of raw, anchor SFT, anchor SDPO and the mixture. Probability gains must remain nonnegative in every rotation stratum and leave-one-source-user-out aggregate. It must also beat each prespecified cheap readout by more than .005 on both probability metrics. Otherwise there is no basis to advance a neural-superiority claim. These are practical development routing thresholds; they are not statistical equivalence tests or validated neural power guarantees. A model-free stress audit invokes the complete rule in 1,200 artificial studies with task and source-user dependence. With no systematic residual advantage, 19 of 200 high-user-noise studies pass, versus zero of 200 low-user-noise studies. These illustrative routing frequencies are not calibrated false-positive rates for the neural experiment. They reinforce the need for independent replication and a separately specified confirmatory analysis.

Dependence, verification and stopping. Average rotations within base; retain source-user dependence in sensitivity analysis. Check the exact expected IDs and semantic option mapping, rather than accepting any complete-looking table. Paired per-base option-order ranges and disagreement are required because global means can cancel opposing order effects. The no-update comparator retains the mass and probability-range requirements but is exempt from an argmax-disagreement gate: a uniform stochastic distribution is a valid baseline despite arbitrary discrete ties. Its disagreement is still reported. Record effective configurations, source and environment hashes, resolved model/tokenizer and weight receipts, initialization and optimizer states, raw logits, tokens and elapsed work. An offline verifier proves only its numerical and provenance checks, not that it independently reran the model.

A qualified negative parks this estimator at this budget. Interface failure permits at most one isolated prospective repair. A pass permits independent seeds, matched-compute controls and a second family, followed by fair published baselines and one separately frozen confirmation. An identity in which delayed and immediate text coincide is checked on CPU; a redundant 35-update training arm is not counted as a transition experiment. No stage automatically turns development success into a submission decision.

7 EXTERNAL MEASUREMENT AND REMAINING EVIDENCE

A delayed response is an informative instrument only under an argued measurement model. A credible external study needs randomized exposure, comparable immediate and delayed measurements of the same declared construct, a neutral comparison, repeated-measure stability or reactivity controls, and prespecified delay and attrition handling. A new elicitation context may reveal a different construct or cause a new transition. Repeated interaction can confound retention of an earlier change with continued treatment.

Public data offer a narrower operational route. We inspected the public release for [Costello et al. \(2026\)](#), which randomizes a conversation about immigration versus smartphones. Its current processed cohort contains 1,108 records; an older public follow-up export uniquely matches 750 of them, with 383 control and 367 treatment respondents. All eleven immediate and follow-up item definitions match after whitespace normalization. The recorded elapsed time from initial-study end to follow-up start is approximately 40–49 days (median 45.54), whereas the source manuscript states a 35-day mean. This unresolved metadata discrepancy must remain visible in a reconstruction. These are availability and linkage checks, not estimated response effects.

Such data can support a separately specified secondary analysis of later *reported attitudes*. They do not supply latent preference ground truth, randomized measurement-reactivity controls, or an empirical bridge to the action-induced target in Equation 1. Roughly one third of the processed cohort

lacks follow-up, and postassignment exclusions prevent an automatic intention-to-treat interpretation. A prospective analysis therefore needs an explicit population, item-scoring rule and bounded-outcome attrition sensitivity. The previously considered conspiracy-belief source (Costello et al., 2024) has a 2026 editorial expression of concern concerning its data pipeline (Thorp, 2026); we do not treat its original public release as a clean replication route. No substantive human-response endpoint has been computed for this draft.

The intended contribution remains conditional on a consequential learning result and an honest external estimand. The strongest unresolved objection is that the construction supplies better labels by assumption and applies a known estimator. Passing a proxy-gradient gate, adding labels unavailable to baselines, or selecting a favorable parser does not resolve that objection.

8 DISCUSSION AND LIMITATIONS

The exact construction separates two latent mechanisms with identical immediate observations. Its value is the explicit measurement and deployment assumptions, not a claim that all personalization is nonidentifiable. The matrix recovery argument and residual estimator are standard, and the finite-state evidence already shows a simpler estimator can be better for decisions. No claim of new general causal identification or general policy superiority follows.

The proposed neural screen measures fixed-label prediction on an observed cohort. It cannot validate Equation 1, identify human persistence, or demonstrate welfare improvement. Its full decision rule has undergone model-free stress checks and still needs empirical replication; constructed Gaussian power for one marginal endpoint cannot qualify the whole study. The same development data must not be used to repeatedly repair a scientific failure until it passes. Submission is contingent on completing the missing evidence, not on the existence of this draft.

REPRODUCIBILITY STATEMENT

The exact proofs are in the appendix. Displayed finite-state aggregates and Figure 1 are generated from preserved cell rows with input hashes. The corrected audit distinguishes formal claims, developmental simulations, invalid assays, qualified negatives and unrun stages. A prospective protocol and source freeze govern the reduced screen; immutable evidence and a separate offline verifier are required for any eventual result. No anonymous artifact URL is claimed before a concrete release exists.

ETHICS STATEMENT

Matching a post-action preference is not assumed to be welfare or consent. Preference measurement can itself influence users. No new human study has been conducted for this draft; any collection requires an appropriate consent, review, privacy and follow-up plan before recruitment.

AI USE STATEMENT

AI systems assisted with hypothesis development, theoretical formulations and proof checking, literature search, experimental design, code and test generation, synthetic-data analysis, scientific interpretation and manuscript drafting. This assistance includes substantive research contributions, not only language editing. The human authors must verify all claims, sources and artifacts before submission and remain responsible for them.

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A IDENTIFICATION, RESTRICTED DECISION RISK, AND MEASUREMENT ASSUMPTIONS

This appendix gives an exact counterexample and the assumptions under which additional measurements resolve it. The example is a construction, not an estimated model of human preferences. Its decision objective is one explicit choice among objectives for influenceable preferences (Carroll et al., 2024). Neither the counterexample nor the sampling identity below establishes a new general estimation method, human prevalence, or a neural learning result.

A.1 TWO MECHANISMS WITH A COMMON OBSERVATION LAW

Let a user’s initial binary state be $Z_0 \sim \text{Bernoulli}(p)$. A logging policy draws $A \in \{0, 1\}$ with $\Pr(A = 1 \mid Z_0 = z) = e_z$. The post-action state is (P_1, S_1) , where P_1 is the persistent component and S_1 is an expressed stance. The initial state is (Z_0, Z_0) , and the common, action-independent observation function is $O = S_1$. Fix copying probabilities $c_0, c_1 \in [0, 1]$. If $A = Z_0$, both mechanisms leave the state unchanged. If $A \neq Z_0$, the expression mechanism E and the transition mechanism T use

$$E : (P_1, S_1) = \begin{cases} (Z_0, A), & \text{with probability } c_A, \\ (Z_0, Z_0), & \text{with probability } 1 - c_A; \end{cases} \quad (4)$$

$$T : (P_1, S_1) = \begin{cases} (A, A), & \text{with probability } c_A, \\ (Z_0, Z_0), & \text{with probability } 1 - c_A. \end{cases} \quad (5)$$

The labels “persistent” and “expression” are part of the construction. They do not turn a recorded utterance into an independently measured state.

Proposition 1 (immediate observational equivalence). For any p, e_0, e_1, c_0, c_1 , the joint law of (Z_0, A, O) is identical under E and T . This remains true even if the historical log records Z_0 without error and the action has randomized overlap.

Proof. When $a = z$, $\Pr(O = z \mid Z_0 = z, A = a) = 1$ in both mechanisms. When $a \neq z$, both give $\Pr(O = a \mid Z_0 = z, A = a) = c_a$ and $\Pr(O = z \mid Z_0 = z, A = a) = 1 - c_a$. Multiplying by the same initial-state and logging probabilities proves equality of every joint atom. Hence the total variation distance is zero. Any independent sample of historical logs, and any statistic or randomized post-processing of that sample, has the same law in the two worlds. This is equality by construction, not a test passed by human data. $\square$

For a fresh user drawn independently from the same initial-state population, define the value of an intervention that always chooses action a by

$$V_m(a) = \Pr_m\{a = P_1(a)\}, \quad m \in \{E, T\}. \quad (6)$$

The evaluated action itself may change the state against which it is scored. This objective rewards post-action matching; it is not a definition of welfare or a requirement to preserve the initial state. Direct evaluation gives

$$(V_E(0), V_E(1)) = (1 - p, p), \quad (7)$$

$$(V_T(0), V_T(1)) = (1 - p + pc_0, p + (1 - p)c_1). \quad (8)$$

For the exact witness $p = 3/5$, $c_0 = 1$, $c_1 = 0$, these values are

$$(V_E(0), V_E(1)) = (2/5, 3/5), \quad (V_T(0), V_T(1)) = (1, 3/5). \quad (9)$$

Thus action 1 is uniquely optimal among constant actions in E , while action 0 is uniquely optimal in T . With $e_0 = e_1 = 1/2$, the nonzero immediate-log atoms (z, a, o) have masses

$$(0, 0, 0) : 1/5, \quad (0, 1, 0) : 1/5, \quad (1, 0, 0) : 3/10, \quad (1, 1, 1) : 3/10. \quad (10)$$

The same equivalence also holds for the separately archived logging design $(e_0, e_1) = (7/20, 7/10)$; the logging designs must not be interchanged when reporting a delayed-log total variation distance.

A.2 THE CORRECTED EXPECTED-RISK MINIMAX STATEMENT

A learner receives n independent historical immediate logs D_n and may use independent randomization U . It outputs one action $\delta(D_n, U)$ for a fresh anonymous user. The fresh user's Z_0 is unavailable at deployment and independent of D_n, U . The comparison class contains the two constant actions. Define expected regret, averaging over training data and learner randomization, by

$$R_m(\delta) = \max_{a \in \{0,1\}} V_m(a) - \mathbb{E}_m[V_m(\delta(D_n, U))]. \quad (11)$$

This restriction is essential: it is not a lower bound for all personalized policies with access to the current user's state.

Proposition 2 (two-point minimax risk). For the witness in Eq. (9), for every $n \geq 0$,

$$\inf_{\delta} \max\{R_E(\delta), R_T(\delta)\} = \frac{2}{15}, \quad (12)$$

where the infimum allows randomized learners. It is attained by ignoring the data and choosing action 1 with probability $1/3$.

Proof. By Proposition 1, $q = \Pr_m\{\delta(D_n, U) = 1\}$ is the same under both mechanisms. Equation (9) therefore gives

$$R_E(\delta) = \frac{1}{5}(1 - q), \quad R_T(\delta) = \frac{2}{5}q. \quad (13)$$

If $q \leq 1/3$, the first expression is at least $2/15$; if $q \geq 1/3$, the second is at least $2/15$. At $q = 1/3$ they are equal. The data-independent randomized rule attains this probability for every sample size. $\square$

Correction: deterministic data dependence is not a fixed action. The minimum worst-case regret over the two *fixed* deterministic actions is $1/5$. The stronger assertion that every deterministic learning algorithm has expected worst-case regret at least $1/5$ is false. For $n \geq 1$, consider the deterministic rule

$$\delta(D_n) = 1\{Z_{0,1} = 0\}, \quad (14)$$

where $Z_{0,1}$ is the first historical user's initial state. In either world, $q = 2/5$. Its two expected regrets are $3/25$ and $4/25$, with maximum $4/25 < 1/5$. The random historical sample supplies randomness even though the algorithm has no random seed. Proposition 2 remains a lower bound for this algorithm; no claim that the deterministic subclass attains $2/15$ at each fixed finite n is needed.

Observed-current-state escape. If the fresh user's Z_0 is observed before acting, the policy $A = Z_0$ causes no copying branch and has $A = P_1 = Z_0$ with probability one under both mechanisms. Its value is one and its regret against the optimal policy in that enlarged class is zero. Historical mechanism nonidentification survives, but the restricted anonymous-user decision lower bound no longer applies. Likewise, the witness alone does not establish impossibility for an unknown longitudinal system with other informative interventions or observations.

A.3 WHAT AN ADDITIONAL MEASUREMENT MUST IDENTIFY

In the binary construction, define an ideal neutral probe by the state map $(P_1, S_1) \mapsto (P_1, P_1)$ followed by the same emission function. Its output is $B = P_1$. The only immediate-log atom whose delayed outcome differs between the witness worlds is $(Z_0, A, O) = (1, 0, 0)$: it has $B = 1$ in E and $B = 0$ in T . Consequently, the total variation distance between the extended laws of (Z_0, A, O, B) is $3/10$ under balanced randomized logging, and $(3/5)(1 - 7/10) = 9/50$ under the archived state-dependent logging design. The probe identifies these two stipulated worlds because its exclusion and measurement properties were specified, not because time elapsed.

The following finite-state statement makes the measurement requirement explicit. Let Z_0 and P_1 take values in $\{1, \dots, r\}$ and let $T_a(z, j) = \Pr(P_1 = j \mid Z_0 = z, \text{do}(A = a))$. Assume that:

1. the baseline states being conditioned on are observed without error, have positive probability, and the logged action is randomized conditional on these states with positive probability for each relevant action;
2. the probe does not change P_1 , which remains stable over the specified horizon; and its known measurement law is the row-stochastic matrix $M_{jb} = \Pr(B = b \mid P_1 = j)$, with no remaining direct dependence on action, baseline, or transient expression conditional on P_1 ;
3. the full-population conditional probe distributions are observed or recoverable under an explicitly justified sampling and response model.

Randomizing invitations to a delayed survey does not by itself justify the third condition when actual response depends on the unobserved outcome.

Proposition 3 (known-emission identification). Writing $Q_a(z, b) = \Pr(B = b \mid Z_0 = z, A = a)$, these assumptions give $Q_a = T_a M$. If M has full row rank r , then

$$T_a = Q_a M^+, \quad \|\hat{T}_a - T_a\|_F \leq \|\hat{Q}_a - Q_a\|_F \|M^+\|_2, \quad \hat{T}_a = \hat{Q}_a M^+, \quad (15)$$

where M^+ is its Moore–Penrose inverse. The error bound assumes M is known exactly; it does not account for estimating or misspecifying M .

Proof. Conditioning on P_1 gives $Q_a = T_a M$. Full row rank implies $MM^+ = I_r$, so right multiplication identifies T_a . Submultiplicativity of the Frobenius and spectral norms proves the error bound. $\square$

Small singular values can make finite-sample inversion unstable. Full row rank is a sufficient condition here, not a claim that every target functional requires full transition identification. Conversely, for

$$M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}, \quad T^{(1)} = I_3, \quad T^{(2)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad (16)$$

$T^{(1)}M = T^{(2)}M$ although $T^{(1)} \neq T^{(2)}$. Thus additional probe samples alone do not fix a measurement channel that aliases these states. An unknown emission matrix also leaves a factorization problem; this theorem does not learn or validate it from arbitrary conversational logs.

A.4 THE STANDARD RESIDUAL ESTIMATOR AND ITS LIMITS

For one randomized action stratum, let the finite population of N logged units have immediate agreement scores $O_i \in [0, 1]$ and delayed scores $B_i \in [0, 1]$. The target here is the finite-population mean $\bar{B}_N = N^{-1} \sum_i B_i$. Let S be a uniformly selected subset of k units, sampled without replacement, on which both scores are observed. Then the difference estimator is

$$\hat{\mu}_{\text{diff}} = \bar{O}_N + \frac{1}{k} \sum_{i \in S} (B_i - O_i). \quad (17)$$

It is standard model-assisted estimation, closely related to the mean estimator in prediction-powered inference (Angelopoulos et al., 2023; Mozer, 2026).

Proposition 4 (design expectation). Conditional on the complete finite population, $\mathbb{E}_S[\hat{\mu}_{\text{diff}}] = \bar{B}_N$. More generally, if R_i indicates that unit i is measured and its positive inclusion probability $\pi_i = \Pr(R_i = 1 \mid \{O_j, B_j\}_{j=1}^N)$ is known, then

$$\hat{\mu}_{\text{HT-diff}} = \frac{1}{N} \sum_{i=1}^N \left[O_i + \frac{R_i}{\pi_i} (B_i - O_i) \right] \quad (18)$$

has the same conditional expectation. Dependence among the R_i does not change this expectation identity, although it matters for variance.

Proof. In the first design, $\mathbb{E}[\mathbf{1}\{i \in S\}] = k/N$; linearity gives $\mathbb{E}[k^{-1} \sum_{i \in S} (B_i - O_i)] = N^{-1} \sum_i (B_i - O_i)$. In the second design, $\mathbb{E}[R_i/\pi_i] = 1$ for each unit. Adding $\bar{O}_N$ yields $\bar{B}_N$ in each case. $\square$

For a superpopulation response model using observed covariates W , a corresponding sufficient condition is $\mathbb{E}[R \mid B, O, W] = \pi(W) > 0$ with known $\pi(W)$. A randomized invitation probability is not the full measurement-inclusion probability if nonresponse is selective. Plug-in estimated weights, self-normalization, outcome-dependent selection with unknown probabilities, and fitting a proxy on the same sampled labels require their own analysis; Proposition 4 does not supply a blanket finite-sample unbiasedness guarantee for them.

Clipping changes the expectation. The archived first finite-state implementation clips $\hat{\mu}_{\text{diff}}$ to $[0, 1]$. This can reduce squared error for a target in $[0, 1]$, but does not preserve unbiasedness. For the finite population $N = 2, k = 1, (O_1, O_2) = (1, 0)$ and $(B_1, B_2) = (0, 0)$, Eq. (17) equals $-1/2$ or $1/2$ with equal probability. Its expectation is the true mean zero. After clipping, the expectation is $1/4$. This counterexample concerns estimator expectation; it does not invalidate a properly reported Monte Carlo regret comparison.

Unbiasedness does not imply a decision advantage. For independent identically distributed pairs with finite second moments, the same k uniformly selected labels from N pairs, and coefficient one, direct covariance expansion gives

$$\text{Var}(\hat{\mu}_{\text{diff}}) = \frac{\text{Var}(B)}{k} + \left(\frac{1}{k} - \frac{1}{N} \right) [\text{Var}(O) - 2 \text{Cov}(B, O)]. \quad (19)$$

Here $\text{Var}(\bar{O}_N) = \text{Var}(O)/N$, $\text{Var}(\bar{B} - \bar{O}_S) = \text{Var}(B - O)/k$, and $\text{Cov}(\bar{O}_N, \bar{B} - \bar{O}_S) = \text{Cov}(O, B - O)/N$, which proves the formula. Thus the estimator need not beat the ordinary mean of the same k delayed labels. Smaller estimation error also need not change which action is selected, and a biased estimator can have smaller decision regret on a particular population. All same-label anchor means and simple shrinkage/mixture baselines therefore belong in a method comparison. In the constructed truthful-transition world $B_i = O_i$ for every pair, so its correction is identically zero; observing this equality in a replay is an algebra check, not an independent empirical discovery about human behavior.

A.5 THE EMPIRICAL TARGET IS A SEPARATE ESTIMAND

The constructed EndoPAHF assay assigns a historical interaction an immediate label and a stipulated delayed label. It then scores a later model answer against that fixed label. No state-transition kernel is applied to the *evaluated* later answer. Its expected correct-label score has the form

$$U_B(\theta) = \mathbb{E}_{(X, B) \sim \mathcal{D}} [\pi_\theta(B \mid X)], \quad (20)$$

with a fixed constructed distribution $\mathcal{D}$. If π_θ is a full-vocabulary next-token distribution, all tokens other than the native correct answer token receive reward zero; a constrained option-normalized policy instead defines a different decision rule and must be labeled as such. Neither version equals Eq. (6) merely because the label is called delayed. Established users appearing in both learning and evaluation also do not become unseen users when the query prefix changes.

Replacing (B_i, O_i) in the residual identity by fixed, differentiable delayed and immediate losses can identify the corresponding sampled loss or its gradient when the same sampling conditions hold and differentiation commutes with expectation. It does not prove that this gradient improves Eq. (20), let alone an action-dependent human state objective. Feedback generated from the learner’s own policy adds distributional derivatives not supplied by a fixed-target identity. The distinction between likelihood, held-out feedback gradients, and value is already central to Nguyen et al. (2026).

Finally, an immediate-versus-delayed human survey study can identify effects on a specified measured response under randomization and justified attrition assumptions. It cannot establish $B = P_1$, a known full-rank M , absence of measurement reactivity, or a uniquely correct welfare objective simply by having a follow-up date. Those are separate validation and normative tasks.

B COMPLETE CONSTRUCTED COMPARISON

Selection	Immediate	Anchors	Residual	Anchor HT	Residual IPW
0.00	0.34567308	0.00380385	0.00222308	0.00676538	0.00417692
0.25	0.34592308	0.00040000	0.00196154	0.00596923	0.00305385
0.50	0.34596154	0.00001538	0.00211154	0.00456923	0.00280385
0.75	0.34582692	0.00000769	0.00223846	0.00418846	0.00218077

The selected subsets match the original declared design predicates. Their averages do not estimate population prevalence. Complete raw cells and unselected diagnostics remain in their historical evidence roots.